

Development of Autonomous UAV Quadcopters using Pixhawk Controller and Its Flight Data Acquisition

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Abstract— Unmanned Aerial Vehicle (UAV) quadcopter is a copter with four driving motors. It is widely used in various applications including agriculture because the design is simple and the concept of motion is easy to understand. However, currently, the widely sold quadcopters in the Indonesian market are quadcopters that are controlled manually using remote control. In this work, we developed two autonomous UAV quadcopters that can fly autonomously according to two predetermined trajectories, *i.e.*, rectangular and circular trajectories. This preliminary work aims to support various quadcopter applications for smart and precision farming on fix agricultural lands. The trajectories are predetermined in accordance to the landforms and the quadcopters are expected to follow these trajectories autonomously. This study has succeeded in building two autonomous quadcopter with different specifications: quadcopter type F450 and quadcopter type F330. The experimental testing results show that the average latitude and longitude mean squared error of the quadcopter flight is 6.79×10^{-10} for the F450 quadcopter and 1.45×10^{-8} for the F330 quadcopter. The flight sensors data obtained in this study is available to be used for research and educational purposes. Please contact the authors for access to the UAV flight data.

Keywords—autonomous, UAV, quadcopter, pixhawk, flight data.

I. INTRODUCTION

Unmanned Aerial Vehicle (UAV) is a vehicle that can fly without the existence of humans / pilots on-board. It can be controlled remotely using a remote control or fly autonomously using an autopilot system. Due to their simple design, multi-rotor UAVs have been applied in various fields [1], such as agricultural [2], [3], [4], military [5], [6], aerial photography [7], [8], search and rescue mission [9], [10], and many other fields. Multi-rotor UAVs can have three, four, six, or eight rotors, which is known as tricopter, quadcopter, hexacopter, and octocopter respectively. Quadcopter is a type of multi-copter that is widely used among others because it has a less complex stabilization mechanism than tricopter. Furthermore, it is cheaper than hexacopter and octocopter since it is built with fewer parts. Due to these advantages, a study on using ANN for the control of quadcopter has been reported in [11] based on the work reported in [12].

Nowadays, the use of quadcopter in agricultural field is increasing, especially for tasks that do not require the UAV to lift a heavy load, such as for surveillance [13], plantation

monitoring and yield estimation [14], real-time crop monitoring [15], and livestock counting [16]. Some recent applications of UAV in agriculture are the use of UAV for *Ganoderma* identification on oil palm tree [17] and *Gemini* virus attack analysis in chili field [18]. However, despite the fact that current researches have been shifting to the design of autonomous quadcopters [19], in Indonesia, most quadcopters that can be found on the market are those that are still manually controlled by using remote control. This requires additional human labor with special expertise and is not in accordance with the concept of smart farming, *i.e.*, agricultural productivity is to be increased while minimizing the use of resources. In any smart and precision farming applications, the use of autonomous quadcopter becomes a more promising choice. This preliminary study aims to build two autonomous quadcopters using F450 and F330 frames, to be used on fix agricultural lands.

This paper is organized in five sections. Section I introduces the general problem and aim of the study. Section II describes the methodology in developing the two autonomous quadcopters. Then, Section III elaborates the development of F450 and F330 quadcopters in details, from the required components, system block diagram, components assembly and placement, trajectory generation and flight data acquisition. Section IV presents and analyses the experimental autonomous-flight testing results for both quadcopters. Finally, the conclusion of this work and the future research directions are presented in Section V.

II. METHODOLOGY

This study is divided into seven stages as depicted in Fig. 1. The work began with the assembly of quadcopter components. Next, the functionality of each quadcopter components is tested to check if all the electronic components can function well and can communicate with the other components. The next process is system configuration and calibration of the sensors, until the default software provides successful notifications. Then, the automatic flight trajectory is generated using Mission Planner software and embedded to the Pixhawk controller. Utilizing this embedded trajectory, the quadcopter flies autonomously, and the flight data is retrieved from the sensors. The flight data is then processed to be filtered and synchronized, as described in detail in [20]. Next, an evaluation is conducted by calculating the mean squared error between the desired latitude and longitude waypoints and the real trajectory from the quadcopter flight data.

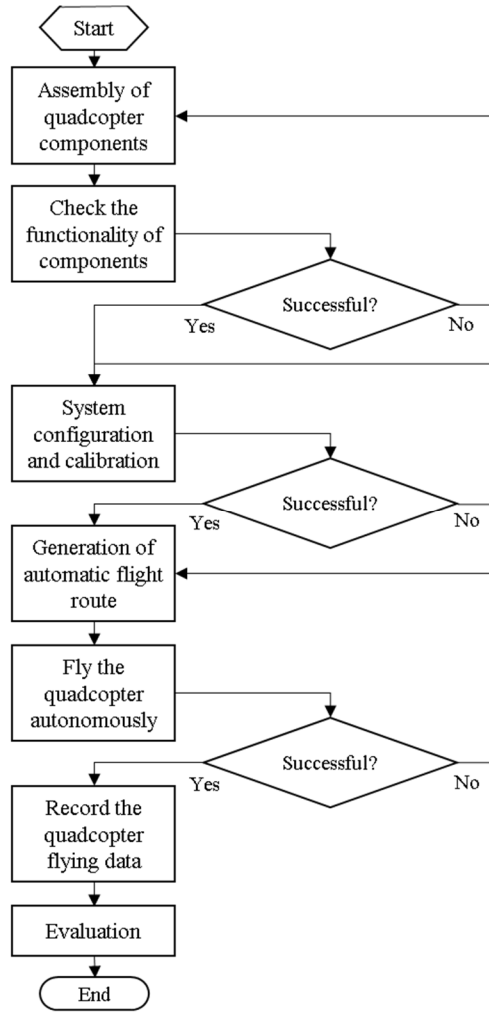


Fig. 1. Research methodology

III. AUTONOMOUS QUADCOPTERS DEVELOPMENT

This study developed two quadcopters, one with a diagonal dimension of 450 mm (F450 frame) and one with a diagonal dimension of 330 mm (F330 frame). The components of the two quadcopters are listed in Table I.

Fig. 2 shows the design of the two quadcopters in block diagram. In the figure, it can be seen that components such as remote control (RC) receiver, radio telemetry, GPS module, safety switch, buzzer, and electronic speed controls (ESCs) are directly connected to Pixhawk controller which is the core of the system. Pixhawk controller has a built-in inertial measurement unit (IMU) module, which has four sensors, *i.e.*, the magnetometer, gyroscope, barometer, and accelerometer. Each motor is connected to an ESC, from which the electric current is obtained from the battery via the power module. The power module divides the electric current into two paths, one is dedicated to power the Pixhawk controller and the other one is used to power the four ESCs. The first electric current will turn on Pixhawk and supply electricity to all other components connected to Pixhawk. Meanwhile, the second current will turn on the ESCs and supply electricity to the corresponding motors.

TABLE I. QUADCOPTER COMPONENTS

	Type F450	Type F330
Frame	F450 quadcopter frame	F330 quadcopter frame
Flight controller	Pixhawk Radiolink	Pixhawk Radiolink
GPS	UBLOX NEO-7M	Radiolink TS100
Radio Control (for safety)	Turnigy TGY 9x Turnigy Rx 9X 8C v2	Radiolink AT9S Rx Radiolink R9DS
Battery	Gens Ace 5000mAh 3S 11.1Volt 45C	Gens Ace 5000mAh 4S 14.8Volt 45C
Electronic Speed Controller	4 ESCs Simonk 30A	4 ESCs Hobbywing 40A
Motor	4 brushless motor sA2212/13T 1000KV	4 brushless motors Sunnysky X2212-9 1400KV
Propeller	4 propeller 1045 (with propeller shield)	4 propeller 7045 (3 blade)
Wireless telemetry for flight data acquisition	2 433 MHz radio telemetry 500mA	2 433 MHz radio telemetry 500mA
Miscellaneous	RC PPM encoder Power module Safety switch Buzzer Pixhawk damping plate GPS folding antenna Quadcopter tall landing gear	OSD telemetry module Radiolink PRM 03 Power module Safety switch Buzzer Pixhawk damping plate GPS folding antenna F330 Quadcopter landing gear

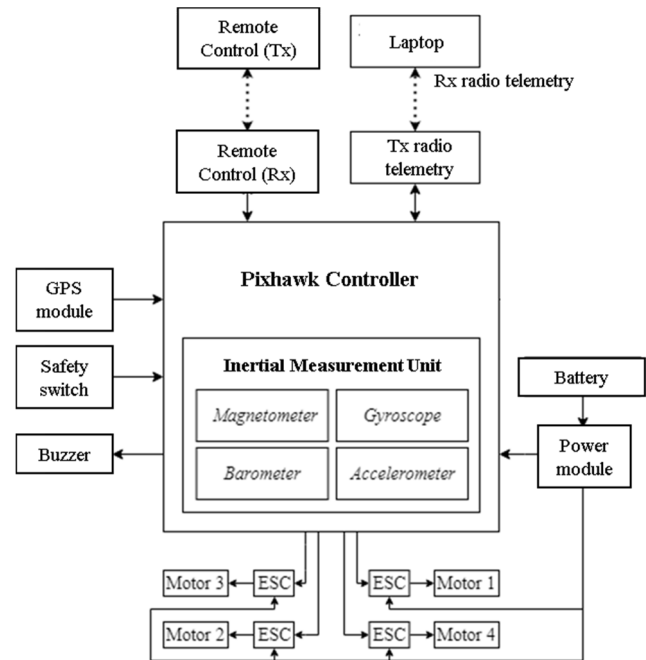


Fig. 2. Block diagram of the developed quadcopters

A. Assembly of Components

The components are assembled according to the block diagram in Fig. 2. Both quadcopter frames, *i.e.*, F450 and F330, have two layers for components in the center of the frame and four arms for motors and propellers. Table II shows the positions of each component in both quadcopters. Fig. 3 and Fig. 4 show the developed quadcopter F450 and quadcopter F330, respectively.

TABLE II. POSITION OF COMPONENTS IN THE QUADCOPTER FRAME

Component	Position	
	Quadcopter F450	Quadcopter F330
Pixhawk	Top layer (center)	Top layer (center)
GPS	Left-front arm (base)	Left-front arm
RC receiver	Bottom layer (rear)	Bottom layer (rear)
Battery	Between top and bottom layer	Hanging on the bottom layer
ESC	Each arm (bottom)	Each arm (bottom)
Motor	Each end of the arm	Each end of the arm
Radio telemetry	Right-rear arm (base)	Right-rear arm (base)
RC PPM encoder	Attached to the RC receiver	-
Power module	Base of the front-left leg	Between top and bottom layer
OSD module	-	Tied (hanging) on the frame
Safety switch	Top layer (left)	Bottom layer (rear)
Buzzer	Left-rear arm (base)	Left-rear arm (base)
Propeller	Installed in accordance with the motor (CW and CCW)	Installed in accordance with the motor (CW and CCW)
Propeller shield	Each end of the frame arm	-
Damping plate	Pixhawk holder on the top layer	Pixhawk holder on the top layer
GPS folding antenna	GPS holder at the base of the front left arm	GPS holder at the base of the front left arm
Landing gear	Bottom of the frame	Bottom of the frame

For quadcopter type F330, the installation of GPS Radiolink TS100 requires some adjustments because its connector is GH1.25, whereas the GPS port in Pixhawk controller is a 6-pin JST.

Quadcopter F450 requires a PPM encoder because the RC receiver in quadcopter F450, *i.e.*, Turnigy Rx 9X 8C v2 can only support 8 channels of PWM signal, one PWM signal per cable. On the other hand, Pixhawk controller cannot receive the signal from RC receiver in the form of one PWM signal per cable. Fortunately, the Pixhawk controller can read a PPM signal, so we add a PPM encoder to convert 8 PWM signals to one PPM signal. The case is different for quadcopter F330 since the utilized RC receiver, *i.e.*, Rx Radiolink R9DS can support the transmission of PPM signal.

Additional on-screen display (OSD) telemetry module is added to quadcopter F330 to check some quadcopter parameters' status in realtime via the remote control screen. Those parameters include speed, rise, throttle, voltage, longitude, latitude, altitude, GPS, RSSI, flight mode, yaw, roll, pitch, and distance.



Fig. 3. Quadcopter F450



Fig. 4. Quadcopter F330

B. Desired Trajectory and Waypoints Generation

The desired quadcopter flight trajectories are generated based on the GPS coordinates in accordance with the shape and boundaries of the agricultural land. In this initial study, to test the two quadcopters, we generate two flight trajectories in the Gymnasium of IPB University. The first trajectory is a set of GPS coordinates that form a rectangular path (Fig. 5) and the second route is a set of GPS coordinates that form a circular path (Fig. 6). Fig. 5 shows that the rectangular trajectory is generated by using five GPS coordinates to represent the center of the field and its four corners. Meanwhile, the circular trajectory in Fig. 6 is generated by using 37 GPS coordinates. Based on the trajectories, four autonomous flight test scenarios are performed as shown in Table III, where each quadcopter is expected to be able to fly autonomously by following the desired trajectories.

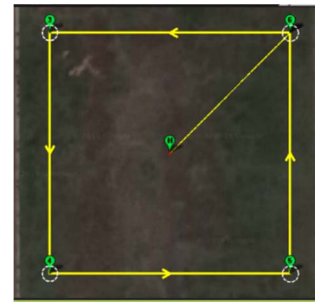


Fig. 5. The generated rectangular trajectory

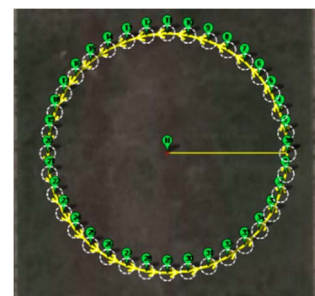


Fig. 6. The generated circular trajectory

TABLE III. QUADCOPTER TESTING SCENARIOS

Scenario	Type	Predetermined trajectory	Flight height
1	F450	Rectangular (60 × 60 m)	10 m
2	F450	Circular (diameter 60 m)	10 m
3	F330	Rectangular (60 × 60 m)	10 m
4	F330	Circular (diameter 60 m)	10 m

C. Autonomous Flight Data Acquisition

The flight experiments were all conducted early in the morning when the weather was good to minimize the interference from the wind, rain, etc. First, the desired flight trajectory is uploaded to the quadcopter using the radio telemetry. Then, the quadcopter carried out this flight mission autonomously. When flying on a mission, various data logs are obtained from the sensors and from the motors, in form of a data-flash log which is stored on a micro SD in the Pixhawk controller. After successfully completing the flight mission, the data-flash log can be easily downloaded to be further analyzed.

The four data-flash logs are processed by taking the required attribute groups based on the ardupilot.org documentation. Some of its attributes are explained in Table IV. The data is synchronized by following the largest sampling time using the C++ program. Detailed explanation on the data acquisition process can be found in [20]. These obtained data can be used for any research and educational purpose by contacting the authors of this paper.

TABLE IV. QUADCOPTER DATA ATTRIBUTES

Attribute	Description
Attitude	
Roll	Actual quadcopter roll angle in degrees (left roll: negative, right roll: positive)
Pitch	Actual quadcopter pitch angle in degrees (forward pitch: negative, back pitch: positive)
Yaw	The actual direction of the quadcopter in degrees (0° = north)
Control, Throttle and altitude information (CTUN)	
Barometer Altitude	Height above ground based on the barometer in meters
Climb Rate	Accelerometer + estimated barometer in cm/s
Global positioning system (GPS)	
Latitude	GPS latitude data
Longitude	GPS longitude data
Horizontal velocity	Quadcopter horizontal speed in m/s.
HDop	Dilution of precision which represent the precision of the acquired GPS data (1.5: good, >2.0 not good).
Global position accuracy (GPA)	
HAcc	Horizontal accuracy as reported by the GPS module in meters.
SAcc	Speed accuracy as reported by the GPS module in meters.
Inertial measurement unit (IMU)	
GyrX	The raw value of the gyro rotation rate in deg/s for the x, y and z axes.
GyrY	
GyrZ	
AccX	The acceleration raw value in m/s² for the x, y and z axes.
AccY	
AccZ	
Magnetometer (Compass)	
MagX	Raw magnetic field values for x, y and z axes.
MagY	
MagZ	
OfsX	Magnetic raw offset value (will only change if the COMPASS_LEARN parameter is 1)
OfsY	
OfsZ	
RC output (RCOUT)	
C1	PWM commands that are sent from the flight controller to the ESC / motors.
C2	
C3	
C4	

IV. RESULTS AND DISCUSSIONS

The real GPS trajectory of each flight experiment data is compared to the desired trajectory (waypoint path) using mean squared error (MSE) as shown in equation (1).

$$MSE(x, y) = \frac{1}{N} \sum_{i=1}^N (x_i - y_i)^2, \quad (1)$$

where x is the longitude and latitude data of the real flight trajectory, y is the longitude and latitude data of the desired trajectory (waypoints) and N is the number of waypoints. Here, we use $N = 37$ for both the rectangular and circular trajectories. For the case of rectangular trajectory, we generate 37 rectangular waypoints from the five GPS coordinates as shown in Fig. 5. Meanwhile, the waypoints for the circular trajectory are directly taken from the 37 GPS coordinates as shown in Fig. 6.

The result of the first experimental test is depicted in Fig. 7. Here, the trajectory of the quadcopter F450 real flight test is compared to the desired rectangular trajectory (waypoints). The obtained longitude MSE for this experiment is 1.13×10^{-10} and the latitude MSE is 5.98×10^{-11} . The result of the second experimental test for quadcopter F450 circular trajectory is shown in Fig. 8. The obtained longitude MSE for this experiment is 7.43×10^{-10} and the latitude MSE is 1.8×10^{-9} . It should be noted that each degree of longitude and latitude is approximately 111.3 km and 110.6 km apart at the equator. Taking these values into consideration, the MSE values equal to 1.40×10^{-6} km², 7.31×10^{-7} km², 9.20×10^{-6} km², and 2.20×10^{-5} km², respectively.

The results of experimental test for quadcopter F330 are shown in Fig. 9 and Fig. 10 for the rectangular and circular trajectories, respectively. The longitude and latitude MSEs for the rectangular testing are 1.18×10^{-10} and 4.42×10^{-11} , whereas that for the circular testing are 1.1×10^{-8} and 4.7×10^{-8} . Taking the equivalence of longitude and latitude degrees to kilometers into consideration, these MSE values equal to approximately 1.49×10^{-6} km², 5.41×10^{-7} km², 1.36×10^{-4} km², and 5.75×10^{-4} km², respectively.

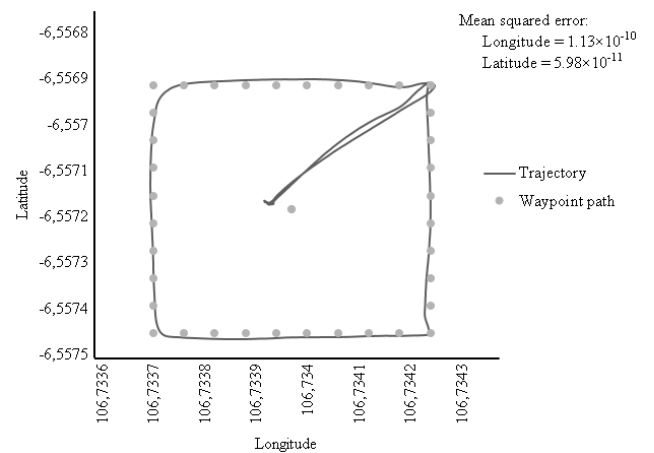


Fig. 7. The experimental testing result of quadcopter F450 – rectangular trajectory mission

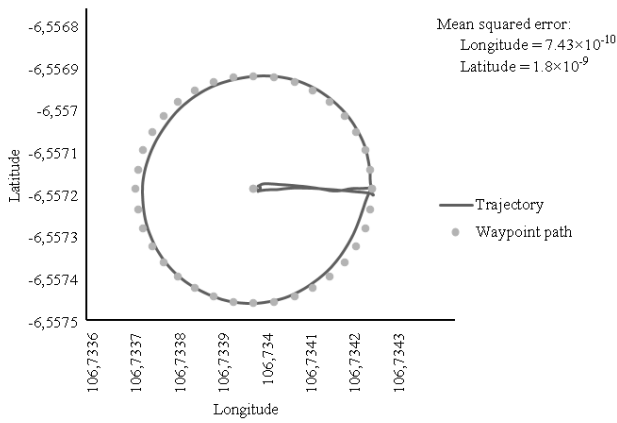


Fig. 8. The experimental testing result of quadcopter F450 – circular trajectory mission

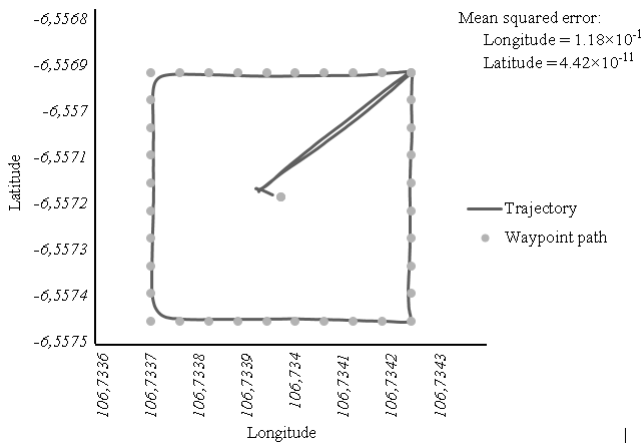


Fig. 9. The experimental testing result of quadcopter F330 – rectangular trajectory mission

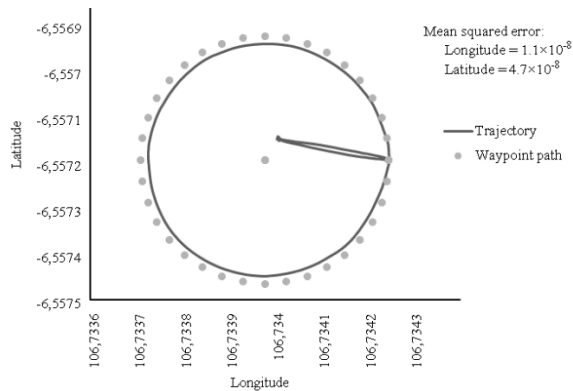


Fig. 10. The experimental testing result of quadcopter F330 – circular trajectory mission

The overall evaluation of the whole experiments are shown in Table V. This results have proven that both quadcopters can fly autonomously by following the desired trajectories. Disturbances such as the light winds may cause some discrepancies on the flight trajectory. These phenomenon are captured by the sensors so it can be used in the next research to further analyze the UAV quadcopter behavior. Other data as listed in Table IV are also available to be used in the next research.

TABLE V. RESULTS OF QUADCOPTER EXPERIMENTAL TESTING

Scenario	MSE	
	Longitude	Latitude
1. Quadcopter F450, rectangular trajectory	1.13×10^{-10}	5.98×10^{-11}
	$1.40 \times 10^{-6} \text{ km}^2$	$7.31 \times 10^{-7} \text{ km}^2$
2. Quadcopter F450, circular trajectory	7.43×10^{-10}	1.8×10^{-9}
	$9.20 \times 10^{-6} \text{ km}^2$	$2.20 \times 10^{-5} \text{ km}^2$
3. Quadcopter F330, rectangular trajectory	1.18×10^{-10}	4.42×10^{-11}
	$1.49 \times 10^{-6} \text{ km}^2$	$5.41 \times 10^{-7} \text{ km}^2$
4. Quadcopter F330, circular trajectory	1.1×10^{-8}	4.7×10^{-8}
	$1.36 \times 10^{-4} \text{ km}^2$	$5.75 \times 10^{-4} \text{ km}^2$

V. CONCLUSION

Autonomous UAV quadcopter type F450 and F330 have been successfully developed. Both quadcopters have been able to carry two experimental testing missions, *i.e.*, autonomous flight tests with rectangular and circular trajectories. The experimental testing results show that the latitude and longitude mean squared error of quadcopter flight is 6.79×10^{-10} for the F450 quadcopter and 1.45×10^{-8} for the F330 quadcopter.

Quadcopter flight data are obtained from various sensors including the gyroscope, accelerometer, magnetometer, barometer, GPS, as well as the pulse width modulation (PWM) signal data. These data can represent all quadcopters' attitudes during flight, *i.e.*, take-off, hover and landing, and are available to be used for further research on quadcopter movement.

The future work of this preliminary study is to implement the two autonomous quadcopters to some specific agricultural tasks, such as surveillance, fertilizing, and pest control. Some additional hardware including camera and actuators are required, depends on the defined task. Image processing algorithm will also be implemented in the future for the purpose of agricultural surveillance.

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